Lab 3 Architecture Justification

Airport Lost Luggage Claim & Tracking Portal

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Architecture selection

We chose **Layered Architecture** for the Airport Lost Luggage Claim & Tracking Portal. Passenger Portal and Agent Console form the presentation layer; Claim Service and Scan Matching Engine form the application layer; Claim & Scan Store, Terminal Scan Adapter, and Notification Service are data and integration components.

Two scenario-specific reasons

1. Passengers file and track claims, while baggage agents investigate and change status. A shared Claim Service applies one claim workflow and consistent rules to both interfaces.
2. The system must reconcile bag-tag scans from connected terminals. Keeping scan ingestion and matching behind interfaces lets terminal feeds change without rewriting claim filing or agent screens.

Architectural styles considered

Style	Assessment for this portal
Layered	Selected: clear UI, claim workflow, and scan/data boundaries.
Microservices	Independent deployment could help at airline scale, but adds network latency and operational complexity to claim matching.
Client-server	Centralized access fits the portal, but by itself gives too little guidance for separating matching, claims, and integrations.

Security advantage

Claim Service checks passenger versus agent permissions before exposing a claim or accepting a status change. Terminal feeds and the data store are not directly reachable from the user interfaces; status changes retain agent ID and timestamp for audit.

Performance benefit

The application layer can query indexed bag-tag and flight records and assemble a multi-leg scan timeline without repeated cross-service calls. This supports the Lab 1 target of a complete timeline response under 200 ms, which still requires load testing.

The diagram labels six provided/required interfaces and shows the claim, scan, status, and notification flows.